VIBHAV MISRA

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SUMMARY

- **M.S. Data Science graduate** with experience in machine learning, data engineering, and business analytics.
- Proficient in building agentic AI workflows **(LangGraph)**, scalable cloud data pipelines **(AWS)**, and predictive **statistical models**.
- Adept at translating complex technical architectures into strategic business insights through **interactive KPI dashboards** and cross-functional stakeholder communication.

EDUCATION

Pace University, Seidenberg School of Computer Science and Information Systems — **New York City, NY**
Master of Science (MS) in Data Science (STEM)| **Honors:** magna cum laude | GPA 3.76 — May 2026

Chandigarh University, Apex Institute of Technology — **India**
Bachelor of Engineering (BE) in Computer Science with a specialization in Artificial Intelligence and Machine Learning — June 2024

TECHNICAL SKILLS

Programming Languages: Python, C++, SQL, R, SAS
Machine Learning & Data: TensorFlow, scikit-learn, XGBoost, Pandas, NumPy, PySpark, Time-Series Forecasting
Databases & Cloud: PostgreSQL, SQLite, AWS (S3, Lambda, EC2), GCP
Generative AI & LLMs: LangGraph, OpenAI API, Agentic Workflows (HITL), RAG, Prompt Engineering
BI & Visualization: Power BI, Tableau, Excel, Matplotlib, Seaborn, Plotly, Streamlit
DevOps & Engineering: Docker, CI/ CD pipelines, Git/ GitHub, ETL/ ELT Processes

WORK EXPERIENCE

Research Assistant | Pace University - Seidenberg School of CSIS | New York, NY — **August 2026 - Present**

- Engineered cross-platform Python regression pipelines to automate anomaly detection in the open-source CStar C++ compiler outputs, significantly reducing manual data validation time.
- Optimized data routing and symbol table management in the virtual machine, improving processing efficiency and system scalability.
- Managed code via Git/ GitHub, ensuring strict data governance and deployment-ready software standards for open-source releases.

Data Science Intern | EdMyst Inc. | Lewes, DE — **June 2025 - September 2025**

- Architected a 2-layer long-term memory framework for an AI coach, integrating 7 user-data sources to enable context-aware, adaptive agentic responses.
- Engineered a Python context-builder for dynamic LLM prompt generation, leveraging 10+ behavioral signals to personalize cross-session AI coaching.
- Developed real-time video ingestion pipelines (AWS S3, WebSockets) processing 94MB files in <30s for continuous model evaluation.
- Automated analytics reporting via Google Apps Script, consolidating 70+ daily data streams into unified KPI dashboards.

PROJECTS

Agentic Text-to-SQL Analytics Assistant | [Live Demo] | [GitHub] | [Docs] — **October 2026**
Tech: Python, LangGraph, OpenAI API, SQLite/ PostgreSQL, Streamlit, Pandas, Matplotlib

- Developed an autonomous Text-to-SQL Agentic RAG workflow using LangGraph, reducing manual query authoring time by 85% while achieving a 92% executable query rate.
- Designed a secure Human-in-the-Loop (HITL) state machine, utilizing agentic self-correction to resolve 90% of syntax errors autonomously prior to real-time CSV extraction via Streamlit.

Global Economic Stress Early-Warning System | [Docs] — **May 2026**
Tech: Python, Pandas, Scikit-learn, FRED API, SHAP explainability, risk-analysis, time-series backtesting

- Aggregated 90+ years of quarterly macroeconomic indicators (FRED API) across G7 nations to build a predictive early-warning classification system.
- Deployed Random Forest and Logistic Regression models balanced via SMOTE and CTGAN, improving recession-event recall by 35% and achieving an 0.85 AUC-ROC, utilizing SHAP to map cross-country risk.

NYC 311 Service Quality & Operational Performance Dashboard | [GitHub] | [Docs] — **February 2026**
Tech: PostgreSQL, SQL, Power BI, DAX, star-schema modeling, KPI analysis, SLA reporting

- Modeled and analyzed 5M+ NYC 311 service requests using a PostgreSQL star schema to evaluate SLA adherence, agency performance, resolution efficiency, and backlog trends.
- Developed an interactive Power BI dashboard for 15+ agencies, identifying operational bottlenecks including resolution times up to 3x the city median and translating findings into improvement recommendations.